

Dr. Kenza KADRI
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[GitHub](#)

RESEARCH INTEREST

My research interests lie at the intersection of cognition, imaging, and computational modelling. I am particularly interested in (1) the neural substrates underlying decision-making and information gathering, (2) how these substrates are disrupted in psychiatric disorders, and (3) how to model these disruptions and link them to underlying neural circuits and behaviour. To address these questions, I employ a combination of real-time fMRI, behavioural paradigms, and computational modelling.

ACADEMIC POSITIONS

Postdoctoral Researcher, Tübingen University (Germany), October 2023 - ...
Principal investigator: Tobias Hauser
Investigating how endogenous fluctuation of dopamine in the Substantia Nigra (SN) and the Ventral Tegmental Area (VTA) impacts decision-making using a real-time functional Magnetic Resonance Imaging (rt-fMRI)

Guest scientist, Max Planck Institute for Biological Cybernetics. (Germany), October 2023

Demonstrator in Research Methods in Practice, Plymouth University (UK) January – May 2023
Module leader: Claire Walsh
Supervising undergraduate students through a 12-week workshop aiming to build an experiment, collect the data and analyse it. The workshop emphasises on reproducible and open science.

Visiting Student, Oxford University (UK) October – December 2021
Principal investigator: Miriam Klein-Flügge
Linking addiction biomarkers to the connectivity profile of striatum subdivisions using resting-state functional MRI (rs-fMRI) as well as questionnaires from the Human Connectome Project database (n=510).

Research Assistant, Plymouth University (UK) February – August 2020
Principal investigator: Elsa Fouragnan
Collection of preliminary behavioural and fMRI data on healthy participants while they learn to assign credit to different environmental stimuli, including faces, cars and tools. Computational models were used to infer key elements of learning, and we used those model estimates to inform behavioural analyses.

Research Assistant, Montreal University (Canada) May – July 2019
Principal investigator: Christian Casanova
Modelling of the cortico-pulvino-cortical network implicated in sight with a spiking neurons simulator. Experimental data analysis (single cell recording) and scientific poster creation.

EDUCATION

PhD in Computational Neuroscience, Plymouth University (UK) 2020 – 2024
Investigated the neural and behavioural bases of impulsivity and substance use in a transdiagnostic framework, using resting-state fMRI and large-scale datasets (Human Connectome Project, n=512), alongside a large online behavioural study (n=300) combining reinforcement learning tasks and psychometric assessments.

MA in Integrative Biology and Physiology (with distinction), Sorbonne University (France) 2018 - 2020
Study and practice of the mechanism implied in the physiological and pathological functions from the cell to the organism. Scientific modeling applied to neuroscience (with Python, R and MATLAB).

BA in Health biology (with distinction), University of Tours (France) 2015 - 2018
Study and practice of fundamental concepts of biology including: Neurobiology of mental illness, Development of Nervous system, Functional and Comparative neuroanatomy, Endocrinology and Behavioral studies, Psychopharmacology.

BA in Psychology (with distinction), University of Tours (France) 2013 - 2016
Theoretical and methodological knowledge in disciplinary fields of psychology (cognitive, developmental, clinical, social, pathological and biological psychology) General education of neuropsychology with courses about imaging, genetics epigenetics of mental illness.

TRAINING & AWARDS

2022	Computational Psychiatry Summer School, Neuroscience School of Advanced Studies (Venice, Italy)
2021	Drugs and the Brain: from laboratory to clinic, Euron Workshop, Summer school
2020	University of Plymouth Studentship (£61,300)
2020	ERASMUS travelling fellowship (Sorbonne University) (2,250 EUR)
2019	Selected for an interdisciplinary tutor in neuroscience (Sorbonne University (Paris) – Ecole Normale Supérieure, Paris)
2019	FSDIE travelling fellowship (Sorbonne University) (1,600 EUR)
2018	University of Montreal Exchange program (Canada) with a fellowship from FSDIE of the University of Tours (3,200 EUR)

PUBLICATION AND MANUSCRIPTS

Kadri, K., Horsch, B., Dome, L., Kakaei, E., Hauser, T. U. (in prep). Endogenous fluctuations in dopaminergic midbrain impact reward learning.

Kadri, K., Mahrukh, R., Scholl, J., Fouragnan, E. (in prep). Linking transdiagnostic biomarkers to credit assignment impairment in humans.

Kadri, K., Marzuki, A., Hauser, T. U. (submitted to Nature Human Behaviour). Dorsomedial prefrontal cortex as an integrative hub for information gathering.

Kadri, K., Scholl, J., Jensen, D., Klein-Flügge, M., Fouragnan, E. (in prep). Toward a new dimensional approach to addiction: Linking addiction markers to the connectivity profiles of striatum subdivisions ([link](#)).

Dome, L., Hazemans, F., **Kadri, K.**, Wagner, B. J., Webb, A., Hauser, T. U. (2025, under review at PLOS CB). CPM: a Python library for theory-driven modelling in computational psychiatry. ([link](#)).

Palmer, T., **Kadri, K.**, Fakra, E., Lutz, A., Scholl, J., Fouragnan, E. (2024). Differential relationship between meditation methods and psychotic-like and mystical experiences. *PLOS ONE*. ([link](#)).

THESIS

Kadri, K., (2024). Exploring the behavioural and neural bases of impulsivity in a transdiagnostic approach relevant for addiction ([link](#))

PRESENTATIONS

Invited talks and presentations

Kadri, K., (2024). Understanding the Impact of Brain Fluctuations on Decision-Making. Max Planck Institute for Human Cognitive and Brain Sciences, Leipzig, Germany

Kadri, K., (2023) Toward a new dimensional approach to Addiction: Linking Addiction transdiagnostic markers to the Connectivity Profiles of the Striatum Subdivisions. Plymouth, UK

Posters

Kadri, K., Horsch, B., Dome, L., Kakei, E., Hauser, T. U. (2026). Endogenous fluctuations in dopaminergic midbrain impact reward learning. SBDM, Paris, France.

Kadri, K., Marzuki, A., Hauser, T. U. (2026). Dorsomedial prefrontal cortex as an integrative hub for information gathering. CPC, Tübingen, Germany.

Kadri, K., Yaakub, S., Scholl, J., Hosking, B., Komarnyckyj, M., Rushworth, M., Klein-Flügge, M., Fouragnan, E. (2021). Towards a new dimensional approach to addiction: linking addiction markers to the connectivity profiles of striatum subdivisions. NeuroFrance, Strasbourg, France.

Kadri, K., Yaakub, S., Scholl, J., Hosking, B., Komarnyckyj, M., Rushworth, M., Klein-Flügge, M., Fouragnan, E. (2021). Towards a new dimensional approach to addiction: linking addiction markers to the connectivity profiles of striatum subdivisions. Tenth Symposium on Biology of Decision-Making, Paris, France.

Kadri, K., Cortes, N., Casanova, C. (2019). The pulvinar activity modulation modifies neural synchronisation through the cortical hierarchy of visual areas. Sorbonne University, Paris, France.

SCIENTIFIC OUTREACH

Computational Properties of the Prefrontal Cortex 2022: Booklet and general organization assistant

The CPPC 7th workshop was held between the 23rd and the 26th of March 2022 in Oxford University (Oxford, UK). I oversaw the creation and design of the booklet. In addition, in collaboration with students from different universities, I helped with the general organisation of the event.

Connectome in Science : Editor in Chief for “New technology” topic; Editor

Association of students' unions, PhD, engineers and researchers that link the different actors of science. The purpose of this union is to share scientific knowledge in a wide range of disciplines! We have 3 major aims: create, share and educate. This growing network is already supported by Sorbonne University.

SKILLS

Computational Modeling

- Reinforcement learning models
- Machine learning: decision tree, SVM, EM algorithm, Logistic regression, semi-supervised and non-supervised algorithms

Programming language and tools

- Language: Python, MATLAB, R, Latex, PCL and SDL (Presentation NeuroObs), bash, Stan (R)
- Version control: Git

Software

- Software: Photoshop, Illustrator, ImageJ, Inkscape, Zotero, Brackets, Mendeley, Presentation, FSL, Workbench, DSI Studio
- Toolboxes: SPM, Psychtoolbox, PsychoPy

- Data collection platforms: Prolific, Qualtrics, REDCap

Neuroimaging

- Magnetic Resonance Imaging (both healthy and clinical populations) including data collection (certified MR operator), pre-processing, analysis and visualisation
- Quantitative MRI (Multi-Parametric Mapping), resting-state and functional MRI
- Real-time fMRI (Turbo Brain Voyager) including real-time denoising (BIOPAC)
- Data analysis:
 - Rs-fMRI: HCP Workbench
 - Event-related analysis: FSL, SPM

Supervision and teaching

- Supervision of three Master students (including two co-supervision) and two bachelor students (University of Plymouth, University of Tübingen)
- Journal Club organisation (University of Tübingen)

MEMBERSHIP

Since 2022
Since 2026

Cognitive Science Society
Early Career Researcher DGPZ

REFERENCES

Prof. Dr. Phil. Tobias Hauser

Professor of Computational Psychiatry
Developmental Computational Psychiatry lab; Department of Psychiatry and Psychotherapy, University of Tübingen, Germany
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Prof. Elsa Fouragnan

Lecturer in Neuroscience
UKRI Future Leader Fellow.
Head of the Neurostimulation lab at BRIC (Brain Research & Imaging Centre), University of Plymouth
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